Functions

Write a piece of code once. Give it inputs. Use its result.

Today we will write our own functions, use functions from libraries, and build a small Pokemon battle game.

Part 1: Writing functions

From a named block of code to a result you can use.

1. **Quick recap**
   Lists, tuples and loops
2. **What is a function?**
   Code we can use again
3. **Defining and calling**
   Write it, then run it
4. **Parameters and arguments**
   Give a function inputs
5. **Return values**
   Get a result back
6. **Execution and scope**
   Follow calls and variable names

Practice: draw a face, track pizza slices, and trace a function.

01 - Quick recap

We can store several values and work through them with a loop.

```
scores = [12, 18, 24]          # A list
person = ("Mina", 20)          # A tuple

total = 0
for score in scores:
    total += score

print(total)                   # 54
print(person[0])               # Mina
```

Today, we will put code like this inside **functions**.

02 - What is a function?

A named piece of code that performs a task.

You already call functions

```
name = input("Your name: ")
print(name)
print(len(name))
```

Function	What it does
`input()`	Reads text and gives it back to us
`print()`	Displays something
`len()`	Gives us the length of something

Some functions take inputs. Some give us a useful result.

Why write our own?

Imagine using the same code in several places.

- Give that task a clear name.
- Call it whenever you need it.
- Change its code in one place instead of many.

```
show_menu()
```

A name such as `show_menu` tells us what the code is for. Next, we will learn how to write that function.

03 - Defining and calling

Defining writes the instructions. Calling runs them.

Anatomy of a function

```
def show_menu():
    print("1. Play")
    print("2. Quit")
```

Part	Meaning
`def`	Start a function definition
`show_menu`	The name we choose
`()`	Where parameters will go
`:`	The indented block starts next
Indented lines	The **body**: instructions indented by four spaces

A definition is not a call

```
def show_menu():
    print("1. Play")
    print("2. Quit")

print("Welcome!")
show_menu()
print("Choose an option.")
```

```
Welcome!
1. Play
2. Quit
Choose an option.
```

Python runs the body only when it reaches `show_menu()`.

Call it again

```
def show_menu():
    print("1. Play")
    print("2. Quit")

show_menu()
show_menu()
```

Each call starts the body again.

Write the definition before the first call runs. Use parentheses to call a function: `show_menu()`, not just `show_menu`.

04 - Parameters and arguments

Run the same instructions with different inputs.

Give a function an input

```
def show_score(score):
    print(f"Score: {score}")

show_score(12)
show_score(30)
```

```
Score: 12
Score: 30
```

Parameter: the name in the definition, such as `score`.

Argument: the value supplied by a call, such as `12`.

More than one input

```
def show_score(name, score):
    print(f"{name}: {score}")

show_score("Mina", 12)
show_score("Sam", 30)
```

Arguments are matched to parameters in order.

You can also use their names:

```
show_score(score=12, name="Mina")
```

These are keyword arguments. The names must match the definition.

A default value

```
def show_score(name, score=0):
    print(f"{name}: {score}")

show_score("Mina")       # Mina: 0
show_score("Sam", 30)    # Sam: 30
```

The default is used only when that argument is omitted.

Put parameters without defaults before parameters with defaults.

Exercise: Draw a face

Write `draw_face(eyes, mouth)`.

- Print the eyes twice, with a space between them.
- Print the mouth on the next line, with one space before it.
- Call it twice with different characters. Choose your own!

For `draw_face("o", "-")`, the output is:

```
o o
 -
```

Folder: `exercises/07-functions/draw-a-face/`

Solution: Draw a face

```
def draw_face(eyes, mouth):
    print(f"{eyes} {eyes}")
    print(f" {mouth}")

draw_face("o", "-")
draw_face("^", "v")
```

```
o o
 -
^ ^
 v
```

The function stays the same; its arguments change the face.

05 - Return values

Give a result back to the code that called the function.

A result we can use

```
def double(number):
    return number * 2

points = double(8)
print(points)               # 16
print(points + 5)           # 21
```

`double(8)` gives back `16`.

The code making the call is the **caller**. It can save the returned value, calculate with it, or use it in a condition.

`print` AND `return` ARE DIFFERENT

Display a value

```
def show_double(number):
    print(number * 2)

result = show_double(8)
print(result)
```

```
16
None
```

Give back a value

```
def double(number):
    return number * 2

result = double(8)
print(result + 5)
```

```
21
```

Without an explicit `return`, a function returns `None`: no useful value.

`return` ENDS THE CALL

```
def ticket_price(age):
    if age < 12:
        return 5
    return 10

print(ticket_price(8))       # 5
print(ticket_price(16))      # 10
```

Once Python reaches a `return`, it goes back to the caller.

For age `8`, the second `return` is never reached.

Exercise: Pizza slices

Write `slices_left(slices, eaten)` that returns the slices remaining.

1. Start with 8 slices and eat 3.
2. Save the returned value in `remaining` and print it.
3. Use `remaining` in another call: eat 2 more slices.
4. Save and print the new result.

```
5
3
```

Assume no one eats more slices than are available.

Folder: `exercises/07-functions/pizza-slices/`

Solution: Pizza slices

```
def slices_left(slices, eaten):
    return slices - eaten

remaining = slices_left(8, 3)
print(remaining)

remaining = slices_left(remaining, 2)
print(remaining)
```

The first returned value becomes an argument to the next call.

If the function only printed the number, we could not use it like this.

06 - How a function runs

Follow a call from start to finish.

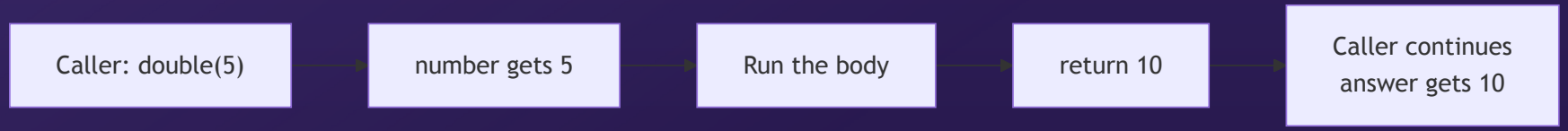
Follow the value

```
def double(number):
    result = number * 2
    return result

answer = double(2 + 3)
print(answer)
```

1. Python defines `double`; it does not run the body yet.
2. It calculates the argument: `2 + 3` gives `5`.
3. It enters the function: the parameter `number` has value `5`.
4. It calculates `result`, then returns `10`.
5. The caller saves `10` in `answer` and prints it.

The path of a call



The caller waits for the function's result before continuing.

If the body reaches its end without a `return`, the result is `None`.

One function can call another

```
def double(number):
    return number * 2

def double_plus_one(number):
    result = double(number)
    return result + 1

answer = double_plus_one(3)
print(answer)               # 7
```

`double_plus_one(3)` calls `double(3)` inside its body.

`double(3)` returns `6`. Then `double_plus_one` adds `1` and returns `7`. Both functions can also be called on their own.

07 - Variable scope

Scope means where a variable name can be used.

Parameters and variables are local

```
def double(number):
    result = number * 2
    return result

answer = double(4)
print(answer)               # 8
```

`number` and `result` are local: their names belong to this function call.

`answer` belongs to the code outside the function. Returning `8` does not make the name `result` available outside.

A local name is not available outside

```
def double(number):
    result = number * 2
    return result

answer = double(4)
print(result)                # NameError
```

The caller received the **value**, not the function's local variable name.

Use `answer` here, not `result`.

Same name, different variables

```
score = 10                  # A global name: outside functions

def show_score():
    score = 3               # A local name: inside this call
    print(score)

show_score()                # 3
print(score)                # 10
```

Assignment inside the function normally creates a local variable. It does not reassign the global `score`.

Each call has its own local names

```
def first_point():
    points = 0
    points += 1
    return points

print(first_point())        # 1
print(first_point())        # 1
```

The second call starts again at `points = 0`.

Local variables are not shared between separate calls.

Reading a global name

```
school = "ReDI"

def show_school():
    print(school)

show_school()                # ReDI
```

A function can read a name defined outside it.

Usually, prefer an explicit input: `show_school(school)`. Parameters and return values make a function's inputs and outputs easier to see.

Trace execution and scope

```
value = 10

def add_two(value):
    result = value + 2
    print(result)
    return result

answer = add_two(1)
print(answer)
print(value)
```

What three lines are printed? Could we use `result` outside the function?

Task and answers are in the lesson’s exercise and solution READMEs.

Solution: Execution and scope

```
3
3
10
```

- Inside the call, the parameter `value` is `1`, so `result` is `3`.
- The caller saves the returned `3` in `answer`.
- The global `value` stays `10`.
- `result` is a local name. Using it outside would cause a `NameError`.

Break

15 minutes

Part 2: Reusing code

Document our functions, use libraries, and build a game.

8 **Docstrings**
Explain a function's purpose

9 **Libraries and imports**
Use code supplied by others

10 **Installing with pip**
Fetch real Pokemon data

11 **Mini project**
A simple Pokemon battle

12 **Homework and questions**
Practise the main ideas

08 - Docstrings

A **docstring** is a short description inside a function. It is the first statement in the body.

```
def double(number):
    """Return twice the supplied number."""
    return number * 2
```

Use triple quotes. Describe what the function does.

For longer functions, also explain their inputs and returned result.

Read a function's help

```
def double(number):
    """Return twice the supplied number."""
    return number * 2

help(double)
```

Python shows the function's name, parameters, and docstring.

Notice `help(double)`, **not** `help(double(4))`. We want help about the function, not about its returned number.

Comments and docstrings

```
def circle_area(radius):
    """Return the area of a circle with this radius."""
    # The calculation goes here.
    pass
```

- A **docstring** tells callers what the function is for.
- A **comment** can explain a particular line of code.
- `pass` is a placeholder that does nothing. Replace it with working code.

You will add docstrings to both functions in the battle project.

09 - Libraries and imports

We do not have to write everything ourselves.

What is a library?

A **library** is reusable code supplied for other programs to use.

A **module** is a unit of Python code we can import, often a `.py` file. A library can contain several modules.

Python includes a **standard library**. It is installed with Python.

Module	Useful for
`math`	Mathematical functions and constants
`random`	Random choices and numbers

Import a module

```
import math

print(math.sqrt(25))        # 5.0
print(round(math.pi, 2))    # 3.14
```

`import math` makes the module available in this program.

The dot means "use this name from that module": `math.sqrt` is the square-root function in `math`.

Import a function by name

```
from math import sqrt

print(sqrt(25))             # 5.0
```

Now we can call `sqrt()` without writing `math.`.

Import only the names you need. Avoid `from math import *`, which makes it harder to see where names came from.

An import can have a shorter name

```
import random as rand

print(rand.randint(1, 6))
```

`as` gives an imported name an **alias**: another name to use here.

It does not install or change the library. Use clear names; an alias is optional.

Random choices and numbers

```python
import random

names = ["pikachu", "bulbasaur", "charmander"]
opponent = random.choice(names)
damage = random.randint(5, 15)

print(opponent)
print(damage)
```

`choice()` picks one item from a non-empty sequence.

`randint(5, 15)` gives an integer from **5 through 15**, including both ends.

The output can change each time we run the program.

Exercise: Circle area

Write `circle_area(radius)` that **returns**:

```
math.pi * radius * radius
```

Import `math`. Call your function with radius `2`. Print its result rounded to two decimal places.

```
12.57
```

Keep the unrounded value inside the function.

Folder: `exercises/07-functions/circle-area/`

Solution: Circle area

```
import math

def circle_area(radius):
    return math.pi * radius * radius

area = circle_area(2)
print(round(area, 2))
```

Python already supplies `math`; we do not need to install it.

10 - Installing with pip

Use a library that does not come with Python.

Install first, then import

pip installs Python packages. A **package** is an installable bundle of code and other files.

We will use `pokebase` to fetch real Pokemon data.

- Install it once in your Python environment.
- Import it in each program that uses it.
- Installation and fetching the data need internet access.

Installing a package does **not** run it in your program. Importing a missing package causes `ModuleNotFoundError`.

Give the project its own packages

A **virtual environment** is a folder containing a separate Python environment. We will name the folder `.venv`.

Open your exercise folder in VS Code and choose **Terminal → New Terminal**.

Windows (PowerShell)

```
py -m venv .venv
```

macOS

```
python3 -m venv .venv
```

Do this once per exercise folder that needs the package.

Install using that environment

Windows (PowerShell)

```
.\.venv\Scripts\python.exe -m pip install pokebase==1.4.1
```

Then run your code:

```
.\.venv\Scripts\python.exe exercise.py
```

macOS

```
.venv/bin/python -m pip install pokebase==1.4.1
```

Then run your code:

```
.venv/bin/python exercise.py
```

`-m pip` runs pip using that Python. No activation command is needed. Do not commit `.venv`; it is not part of your solution.

Fetch data with a library function

```
import pokebase

pokemon = pokebase.pokemon("pikachu")
stats = []
for stat in pokemon.stats:
    stats.append(stat.base_stat)

print(stats)                # [35, 55, 40, 50, 50, 90]
```

The library handles downloading and reading the data for us.

`pokemon.stats` gives us a list of stats. Each `stat.base_stat` is a number: health, attack, defense, special attack, special defense, and speed.

We can loop over that list. **No dictionaries are needed.**

External data can fail

- Check spelling if a name is not found. Try `pikachu` first.
- Check your connection if the data cannot be downloaded.
- Use the same Python environment to install and run the program.

The project starter includes the error-handling code. You do not need to write it yourself.

The checkers use sample data: they work without internet or installed packages. Run the program with `pokebase` to try the real data.

Exercise: Pokemon stats

Fetch Pikachu using `pokebase`.

- Loop over `pokemon.stats`; each number is in `stat.base_stat`.
- Use a loop to add them together. Do not use `sum()`.
- Print the total.

```
320
```

The numbers are external data; the calculation is our own Python code.

Folder: `exercises/07-functions/pokemon-stats/`

Solution: Pokemon stats

```
import pokebase

pokemon = pokebase.pokemon("pikachu")
total = 0
for stat in pokemon.stats:
    total += stat.base_stat

print(total)                # 320
```

The next task reuses this data in a game.

11 - Mini project: Pokemon battle

Your functions make the game work.

The rules

HP means health points.

- Enter any supported Pokemon name.
- The computer chooses its Pokemon from a short list.
- Both start with their real health from the library.
- Press Enter to attack. Each hit does random damage from 5 to 15.
- If the opponent survives, it attacks back.
- The first Pokemon to reach zero health loses.

This is a simplified classroom game, not the official Pokemon battle system.

Folder: `exercises/07-functions/mini-project-pokemon-battle/`

Write both functions

The starter has placeholders, **not function solutions**.

Function	What students implement
`get_hp(name)`	Fetch the Pokemon and return `pokemon.stats[0].base_stat`: its health
`attack(hp, damage)`	Subtract damage and return the remaining health, never below zero

Add a docstring to each function.

The input and error-handling code is already supplied. It asks again for unknown names and stops clearly if the service fails.

Connect your functions to the loop

The starter supplies this structure. Fill the two missing function calls.

```
while player_hp > 0 and opponent_hp > 0:
    input("Press Enter to attack: ")
    damage = random.randint(5, 15)
    # Update opponent_hp using attack(...).
    print(f"{opponent_name} has {opponent_hp} HP.")

    if opponent_hp > 0:
        damage = random.randint(5, 15)
        # Update player_hp using attack(...).
        print(f"{player_name} has {player_hp} HP.")
```

Save each returned result. Do not let a defeated opponent attack back.

Try your game

```
Your Pokemon: pikachu
You: pikachu (35 HP)
Opponent: bulbasaur (45 HP)
Press Enter to attack:
bulbasaur has 33 HP.
pikachu has 28 HP.
```

This is one possible start. The opponent and damage can change.

Check that health never goes below zero and the game ends. Run `check.py` for predictable tests using sample data.

Solution: The two functions

```
def get_hp(name):
    """Fetch a Pokemon and return its starting health."""
    pokemon = pokebase.pokemon(name)
    return pokemon.stats[0].base_stat

def attack(hp, damage):
    """Return health after damage, with a minimum of zero."""
    remaining = hp - damage
    if remaining < 0:
        return 0
    return remaining
```

The lookup and input helpers in the starter stay unchanged.

Solution: The battle loop

```
while player_hp > 0 and opponent_hp > 0:
    input("Press Enter to attack: ")
    damage = random.randint(5, 15)
    opponent_hp = attack(opponent_hp, damage)
    print(f"{opponent_name} has {opponent_hp} HP.")

    if opponent_hp > 0:
        damage = random.randint(5, 15)
        player_hp = attack(player_hp, damage)
        print(f"{player_name} has {player_hp} HP.")
```

The starter's final condition announces a win or loss. The full solved program is in the matching `solutions/` folder.

12 - Homework

Four short tasks. Each has an `exercise.py` and its own `check.py`.

Task	Practise
1. Ticket prices	Parameters, conditions and returning a price
2. High scores	A list parameter and a loop
3. Average score	Returning a calculation for the caller to use
4. Roll a die	Default parameters, a library and a docstring

Folder: `exercises/07-functions/`

Homework 1: Ticket prices

Write `ticket_price(age)`.

- Return `5` if the age is below `12`.
- Otherwise return `10`.
- Print the returned prices for ages `8` and `16`.
- Add those two returned prices and print the total.

```
5
10
15
```

Folder: `homework-01-ticket-prices/`

Solution: Ticket prices

```
def ticket_price(age):
    if age < 12:
        return 5
    return 10

child = ticket_price(8)
adult = ticket_price(16)
print(child)
print(adult)
print(child + adult)
```

Homework 2: High scores

Write `count_high_scores(scores, minimum)`.

Use a loop to count scores **greater than or equal to** `minimum`. Return the count.

Use `[12, 45, 7, 30]` and a minimum of `20`. Print the result.

```
2
```

Folder: `homework-02-high-scores/`

Solution: High scores

```
def count_high_scores(scores, minimum):
    count = 0
    for score in scores:
        if score >= minimum:
            count += 1
    return count

print(count_high_scores([12, 45, 7, 30], 20))
```

Homework 3: Average score

Write `average(scores)`. Assume the list is non-empty.

Use a loop to find the total, then return `total / len(scores)`.

Call it with `[12, 18, 24]`. Print the result. If it is at least `15`, also print `Pass`.

```
18.0
Pass
```

Folder: `homework-03-average-score/`

Solution: Average score

```
def average(scores):
    total = 0
    for score in scores:
        total += score
    return total / len(scores)

result = average([12, 18, 24])
print(result)
if result >= 15:
    print("Pass")
```

Homework 4: Roll a die

Import `random`. Write `roll_die(sides=6)` with a docstring.

Return a random integer from `1` through `sides`, including both ends.

Print one roll using the default and one using `sides=20`.

One possible output:

```
4
17
```

Folder: `homework-04-roll-a-die/`

Solution: Roll a die

```
import random

def roll_die(sides=6):
    """Return a random integer from 1 through sides."""
    return random.randint(1, sides)

print(roll_die())
print(roll_die(sides=20))
```

The output changes. The checker tests the range and both endpoints.

The main ideas

- **Define** a function with `def`; **call** it with parentheses.
- **Parameters** name the inputs; **arguments** supply their values.
- **Return** gives a result to the caller; **print** displays something.
- **Scope** determines where a variable name is available.
- **Docstrings** describe what a function does.
- **Import** reusable code; **pip** installs additional packages.

Use small functions with clear inputs and useful results.

Questions?